

Mid-Semester Progress Update

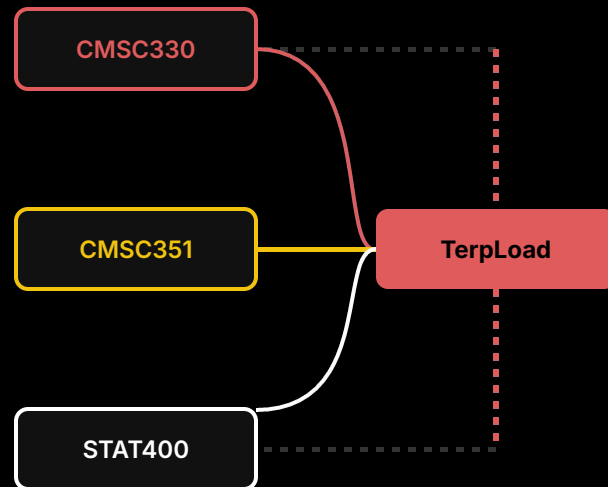
TerpLoad

Understand your semester before the
workload stacks up

**Moataz
Abdelaziz**
Product

**Abhiram
Metuku**
Data & Eval

Sriram Vema
Engineering



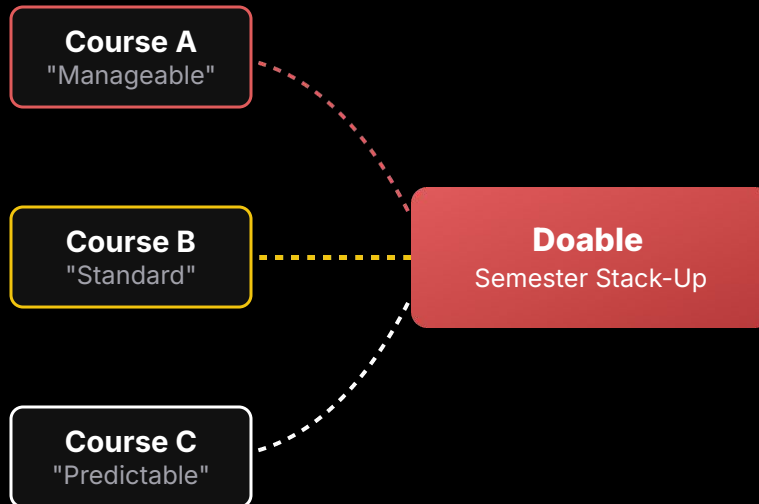
THE STUDENT PROBLEM

The Workload Stack-Up

Target user: UMD CS/STEM students planning combinations of demanding courses.

Individual course planning fails to capture collective reality.

- Students evaluate courses in isolation, missing the compound effect.
- Main problem is that exams, projects, and homework overlap.
- Current planning tools like PlanetTerp, Reddit, and advice provide only isolated, subjective guesses.



DEVELOPMENT STATUS

What We Built So Far

255

Cleaned Reviews

Real PlanetTerp course-review data across seven courses.

30

Review Labeling Pilot

Candidate annotations for project-heavy, exam-heavy, homework-heavy, and time-consuming.

Rule

Preliminary Risk Logic

risk_rules.py produces Low, Medium, or High warnings from preliminary course inputs.

course_id	review_text	rating	review_year	evidence_weight	project_heavy	exam_heavy	homework_heavy	time_consuming
CMSC351	The worst professor	1	2025	1	0	1	0	0
CMSC216	Great professor	5	2025	1	1	0	0	0
CMSC216	All of the projects	5	2025	1	1	0	0	1
CMSC351	My role model. Excellent	5	2025	1	0	0	0	0
CMSC250	CMSC250 is more	3	2025	1	0	0	0	0
CMSC250	telli is a pretty decent	3	2025	1	0	1	0	0
CMSC250	Fall 2025 Pros: -	5	2025	1	0	1	0	0
CMSC216	Actually the worst	1	2025	1	0	1	0	0
CMSC351	Yeah he's a good	5	2025	1	0	1	0	0
CMSC216	One of the best	5	2025	1	1	0	0	1
CMSC351	As I am writing this	2	2025	1	0	1	0	0

Our next steps: Current prototype components work separately. Model training and automatic risk integration are next.

Technical Pipeline Architecture

A structured workflow from multi-source data ingestion to predictive semester risk assessment.

01

Collect Course Reviews

Gather student course reviews from PlanetTerp and other available academic sources.

02

Label Workload Categories

Assign four workload labels (Project, Exam, Homework, Time) using manual and LLM assistance.

03

Text Preprocessing

Clean review text, remove noise, and prepare structured inputs for downstream models.

04

Baseline Model

Convert reviews into TF-IDF vectors and train a classic multi-label Logistic Regression classifier.

05

Deep Learning Model

Fine-tune a DistilBERT transformer model to learn deep contextual representations of review text.

06

Risk Prediction

Aggregate predicted workload labels across selected courses to estimate semester workload risk.

Envisioned End Product (Prototype)

TERPLOAD

Understand your workload and risks for the semester

Select your courses

MSML604

MSML605

MSML606

|



<https://www.figma.com/proto/7EBgoWoZuAX8r0b927VKES/Untitled--Copy-?node-id=0-1&t=IXJoeA0Sokd8OHRo-1>

USER VALIDATION

What We Learned From Students

We interviewed four UMD CS/STEM students, then showed the same students our sample schedule-risk report.

PROBLEM VALIDATION

- Students currently rely on friends, PlanetTerp, and Reddit.
- Difficult course combinations were harder to anticipate than one difficult course.
- Project workload, exam pressure, and time commitment were the most valuable signals.

PROTOTYPE FEEDBACK

4/4

"Students said the sample schedule-risk report would help them plan a semester."

- The report brought scattered workload information into one place.
- Students saw value in evaluating course combinations rather than only individual courses.

HOW FEEDBACK SHAPED

Broad course-advisor idea(our initial idea)



Schedule-combination focus



Four workload signals



Evidence & limited-data warnings

* Small sample; this is early directional feedback, not final usability validation. Plan to ask more people as we refine the mockup

METRIC & PROGRESS

North-Star Metric & Status

Tracking development progress against our student schedule-risk validation goals.

NORTH-STAR METRIC

"Percent of students who say the workload risk explanation helps them plan their schedule."

WHAT IS COMPLETED

- Problem validated with target students
- Data Ingestion pipeline started
- Four-label annotation schema set
- Risk-rule heuristic baseline operational

CURRENT PROGRESS TREND

Raw reviews		903
Cleaned reviews		255
Pilot labels		30
Seed courses		10
Interviews		4

WHAT IS IN PROGRESS

- Label quality and consistency audit
- TF-IDF classical machine learning baseline
- DistilBERT deep learning model skeleton
- End-to-end multi-label evaluation pipeline

The trained classifier and full integrated product are currently in development.

NEXT STEPS

Persevere + Next Steps

Decision: Persevere

Why:

User interviews, course reviews, the report mockup, and pilot labels all support the same pain: students struggle to predict workload stacking across multiple courses.

Feedback Desired:

- Are our four workload labels enough for MVP?
- Is the minimum acceptable demo clear?
- Prioritizing TF-IDF reliability vs. DistilBERT?
- Does the report explain risk clearly?

Wk.8 Expand weak labels + run keyword/TF-IDF baselines

Wk.9 Train DistilBERT + compare models

Wk.10 Aggregate by course and connect risk rules

Wk.11 Build demo and test with users

Wk.12 Freeze features + final evaluation/demo prep

Demo-Day Goal

A student enters courses and receives an **evidence-backed workload-risk report**.